ASSIGNMENT NO 8 ::

1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
d2_96695_sahil> select * from orders o
    -> where o.snum=(select snum from customers where cname='cisneros');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
2 rows in set (0.00 sec)

d2_96695_sahil>
```

7. Extract all the orders of Motika.

```
d2_96695_sahil> select * from orders
    -> where snum=(select snum from salespeople where sname='motika');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
1 row in set (0.00 sec)

d2_96695_sahil>
```

8. Find all the orders of salespeople servicing customers in London.

```
d2_96695_sahil> select * from orders o
    -> where o.snum =all(select snum from customers where city='london');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.00 sec)

d2_96695_sahil>
```

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
d2_96695_sahil> select * from orders o
    -> where amt > any (select amt from orders ord inner join customers c on
 ord.cnum=c.cnum where city='london');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
6 rows in set (0.00 sec)

d2_96695_sahil>
```

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
d2_96695_sahil> select * from customers css
    -> where css.rating >= any(select rating from customers c inner join salespeopl
e sp on c.snum =sp.snum where sp.sname='serres');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)

d2_96695_sahil>
```

11. Select customers who have a greater rating than any other customer in Rome.

```
d2_96695_sahil> select * from customers
    -> where rating > any(select ratig from customers where city='rome');
ERROR 1054 (42S22): Unknown column 'ratig' in 'field list'
d2_96695_sahil> select * from customers
    -> where rating > any(select rating from customers where city='rome');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)

d2_96695_sahil>
```

12. Select all orders that had amounts that were greater that at least one of the orders from '1990-10-04'.

```
d2_96695_sahil> select * from orders
    -> where amt> any(select amt from orders where odate='1990-10-04');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
8 rows in set (0.00 sec)

d2_96695_sahil>
```

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
d2_96695_sahil> select * from orders
    -> where amt < any (select amt from orders ord inner join customers cs on ord.c
num=cs.cnum where city='San Jose');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
8 rows in set (0.00 sec)

d2_96695_sahil>
```

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
d2_96695_sahil> SELECT o.amt,c.cname,c.rating from orders o inner join custo
mers c on o.cnum=c.cnum where amt>(select AVG(amt) from orders);
+---------+---------+--------+
| amt     | cname   | rating |
+---------+---------+--------+
| 5160.45 | Liu     |    200 |
| 9891.88 | Clemens |    100 |
| 4723.00 | Clemens |    100 |
+---------+---------+--------+
3 rows in set (0.01 sec)

d2_96695_sahil>
```

14. Select those customers whose rating are higher than every customer in London.

```
d2_96695_sahil> select * from customers
    -> where rating >all (select rating from customers where city='london');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)

d2_96695_sahil>
```

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
d2_96695_sahil> SELECT s.sname,s.snum,s.city from salespeople s
    -> where s.city != ALL(select c.city from customers c where c.snum = s
.snum);
+---------+------+-----------+
| sname   | snum | city      |
+---------+------+-----------+
| Motika  | 1004 | London    |
| Rifkin  | 1007 | Barcelona |
| Axelrod | 1003 | New York  |
+---------+------+-----------+
3 rows in set (0.01 sec)

d2_96695_sahil>
```

9. Find names and numbers of all salesperson who have more than one customer.

```
d2_96695_sahil> select sname,snum from salespeople
    -> where snum = any(select snum from customers group by snum having count(snum)
>1);
+--------+------+
| sname  | snum |
+--------+------+
| Peel   | 1001 |
| Serres | 1002 |
+--------+------+
2 rows in set (0.00 sec)

d2_96695_sahil>
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
d2_96695_sahil> SELECT s.snum,s.sname,s.city,COUNT(c.cnum) AS customer_count
 from salespeople s
    ->      JOIN customers c ON c.snum = s.snum
    ->      GROUP BY snum ,sname,s.city
    ->      HAVING COUNT(c.cnum)>1;
+------+--------+----------+----------------+
| snum | sname  | city     | customer_count |
+------+--------+----------+----------------+
| 1001 | Peel   | London   |              2 |
| 1002 | Serres | San Jose |              2 |
+------+--------+----------+----------------+
2 rows in set (0.00 sec)
```

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
d2_96695_sahil> SELECT SNUM, SUM(AMT)
     -> FROM ORDERS
     -> GROUP BY SNUM
     -> HAVING SUM(AMT) = (
     ->          SELECT MAX(TOTAL)
     ->          FROM (
     ->                  SELECT SUM(AMT) AS TOTAL
     ->                  FROM ORDERS
     ->                  GROUP BY SNUM
     ->              ) T
     -> );
+------+-----------+
| SNUM | SUM(AMT)  |
+------+-----------+
| 1001 | 15382.07  |
+------+-----------+
1 row in set (0.01 sec)

d2_96695_sahil> |
```